

Sequence:

In Oracle, you can create an autonumber field by using sequences. A sequence is an object in Oracle that is used to generate a number sequence. This can be useful when you need to create a unique number to act as a primary key.

The syntax to create a sequence in Oracle is:

```
CREATE SEQUENCE sequence_name
  MINVALUE value
  MAXVALUE value
  START WITH value
  INCREMENT BY value
  CACHE value;
```

For example:

[illegible]

```
CREATE SEQUENCE dept_seq
  MINVALUE 0
  MAXVALUE 999
  START WITH 1
  INCREMENT BY 1
  CACHE 20;
```

To retrieve the next value in the sequence order, you need to use `nextval`.

For example:

```
supplier_seq.NEXTVAL;
```

This would retrieve the next value from `supplier_seq`. The `nextval` statement needs to be used in a SQL statement. For example:

```
INSERT INTO suppliers
(supplier_id, supplier_name)
VALUES
(supplier seq.NEXTVAL, 'Kraft Foods');
```

Drop Sequence

Once you have created your sequence in Oracle, you might find that you need to remove it from the database.

Syntax

The syntax to drop a sequence in Oracle is:

DROP SEQUENCE sequence name;

With respect to a sequence, the cache option specifies how many sequence values will be stored in memory for faster access.

The downside of creating a sequence with a cache is that if a system failure occurs, all cached sequence values that have not been used, will be "lost". This results in a "gap" in the assigned sequence values. When the system comes back up, Oracle will cache new numbers from where it left off in the sequence, ignoring the so called "lost" sequence values.

To recover the lost sequence values, you can always execute an ALTER SEQUENCE command to reset the counter to the correct value.

Nocache means that none of the sequence values are stored in memory.

This option may sacrifice some performance, however, you should not encounter a gap in the assigned sequence values.

You can change the LASTVALUE for an Oracle sequence, by executing an ALTER SEQUENCE command.

For example, if the last value used by the Oracle sequence was 100 and you would like to reset the sequence to serve 125 as the next value.

You would execute the following commands.

```
ALTER SEQUENCE seq_name  
INCREMENT BY 124;
```

```
SELECT seq_name.nextval FROM dual;
```

```
ALTER SEQUENCE seq_name  
INCREMENT BY 1;
```